■ **Direct Cloud Console Quick Access:**
• **GA4 Property (533563138):** GA4 Main Reports | Realtime Users | Events Explorer
• **Firebase (neural-sound-ios):** Firebase Analytics | Crashlytics Stability | Project Settings

Portfolio Performance Report: Neural Sound App Suite

Author: Principal Mobile Analytics & Monetization Growth Architect
Data Scope: Production Only (Internal tester streams from Bangladesh excluded)
Reporting Period: Yesterday (Oct 9, 2026) & Multi-Horizon Backwards-Looking Review

---

1. Portfolio Executive Summary & Health Dashboard

This section provides a side-by-side comparative analysis of the active mobile applications managed under GA4 Property ID `533563138`.

```
========================================================================================================== PORTFOLIO EXECUTIVE HEALTH DASHBOARD
========================================================================================================== Metric Music Separation iOS (com.neuralsound.mss) Vocal Cleaner (com.neuralSound.audiocleaner)
---------------------------------------------------------------------------------------------------------- Active Users (Yesterday) 71 4 New Users (Yesterday) 19 2 Sessions (Yesterday) 82 9 Engagement Rate (Yesterday) 57.3% 100.0% Avg Session Duration (Yest.) 298.2 seconds (~4.97 mins) 1,737.8 seconds (~28.96 mins) Crash-Free Users (Yesterday) 99.27% 100.0% Revenue (Yesterday) $7.39 $0.00
---------------------------------------------------------------------------------------------------------- Active Users (Past 30D) 2,887 25 New Users (Past 30D) 1,780 25 Sessions (Past 30D) 5,920 152 Engagement Rate (Past 30D) 76.4% 87.5% Product Sales Revenue (30D)* $58.17 $117.64 Multi-Horizon Revenue (30D)* $108.47 $131.09
---------------------------------------------------------------------------------------------------------- Top Selling Product (30D) mss_weekly_sub (8 units, $25.77) 300 minutes consumable (5 units, $96.17) Primary Market (Users) Japan (53.5%), South Korea (19.7%) Israel (25.0%), Nigeria (25.0%)
========================================================================================================== * Note on Revenue Discrepancy: The difference between Itemized Product Sales Revenue and Multi-Horizon Revenue indicates a reporting lag in purchase events vs. subscription state change hooks in GA4, or backdated refund/renewal adjustments.
```

Portfolio-Level Architecture Observations:

1. **The Scale-Monetization Paradox:** *Music Separation iOS* possesses a highly functional acquisition engine (1,780 new users in 30 days) but suffers from poor monetization efficiency (30D ARPU of **$0.037**). Conversely, *Vocal Cleaner* has a highly lucrative monetization layout (30D ARPU of **$5.24** based on multi-horizon figures) but is severely starved of traffic (only 25 active users in 30 days).
2. **SKU Localization Fragmentation:** *Music Separation iOS* shows significant fragmentation in SKU tracking. Dynamic App Store localizations (e.g., ■■, ■■, ■■) are hardcoded into transaction names rather than unified into standardized SKUs (e.g., `mss_weekly_sub`), creating complex reconciliation requirements.
3. **Sandbox Telemetry Contamination:** *Vocal Cleaner* shows a highly suspicious localized purchase pattern on `2026-09-20` in Egypt (comprising 4x 300-minute packs, 2x 60-minute packs, and 1x weekly subscription). This points to an failure to isolate Apple Sandbox / TestFlight transactions from production GA4 event streams.

---

2. Deep-Dive Section: Music Separation iOS

App Header & Health Score

- **App Name:** Music Separation iOS
- **Bundle ID:** `com.neuralsound.mss`
- **GA4 Stream ID:** `14402502632`
- **Overall Health Score: 7.2 / 10**

Rationale: The app shows solid organic acquisition and engagement (76.4% over 30 days). However, it loses points due to:

1. A critical slider telemetry bug generating over 1,700 redundant events per session.
2. A dip in the crash-free user rate to 92.86% on build `1.3.10`.
3. Weak monetization mechanics, heavily relying on weekly subscriptions that have high churn rates.

■ In-App Purchases, Subscriptions & Consumable Breakdown

Product Sales Summary Table (Past 30 Days)

This table displays itemized product sales captured via GA4 purchase events:

Item/Product ID	Product Name (as localized)	Product Type	Units Sold (30D)	Total Revenue ($ USD)	% Revenue Share
`mss_weekly_sub`	■■Subscription (Weekly) Korean)	(Weekly)	5	$16.21	27.87%
`mss_weekly_sub`	■■Subscription (Weekly) Japanese)	(Weekly)	3	$9.56	16.43%
`mss_monthly_sub`	■■Subscription (Monthly) Japanese)	(Monthly)	2	$13.98	24.03%
`mss_monthly_sub`	Monthly Subscription (English)	(Monthly)	1	$8.15	14.01%
`mss_monthly_sub`	■■Subscription (Monthly) Korean)	(Monthly)	1	$7.39	12.70%
`mss_weekly_sub`	Weekly Subscription (English)	(Weekly)	1	$2.88	4.95%
TOTALS			**13**	**$58.17**	**100.00%**

Transaction Timeline Log Table (Chronologically Sorted)

The last 12 transactions captured in the production log:

Date/Timestamp	App Build	Product ID	Product Name	Country	Units	Revenue ($ USD)
2026-10-09	1.3.11	`mss_monthly_sub`	■■	South Korea	1	$7.39
2026-10-06	1.3.11	`mss_monthly_sub`	Monthly	Israel	1	$8.15
2026-10-01	1.3.10	`mss_weekly_sub`	■■	Japan	1	$3.18
2026-09-30	1.3.10	`mss_monthly_sub`	■■	Japan	1	$6.99
2026-09-27	1.3.10	`mss_weekly_sub`	■■	South Korea	1	$3.25

2026-09-25	1.3.10	`mss_weekly_sub`	■■	South Korea	1	$3.22
2026-09-23	1.3.10	`mss_monthly_sub`	■■	Japan	1	$6.99
2026-09-23	1.3.10	`mss_weekly_sub`	■■	Japan	1	$3.18
2026-09-20	1.3.10	`mss_weekly_sub`	Weekly	Egypt	1	$2.88
2026-09-19	1.3.10	`mss_weekly_sub`	■■	Japan	1	$3.21
2026-09-17	1.3.10	`mss_weekly_sub`	■■	South Korea	2	$6.46
2026-09-11	1.3.09	`mss_weekly_sub`	■■	South Korea	1	$3.29

Build Environment & Monetization Analysis:

- **Localization SKU Fragmentation:** There is no centralized product ID reporting. The App Store configuration sends localized language strings as product names (■■, ■■, `Weekly`). **Action:** Implement a mapping layer to convert these to a clean `revenue_sku` parameter (e.g., `com.neuralsound.mss.weekly`) prior to calling `logEvent("purchase")`.
- **Subscription Dynamics:** Consumable options are not available in this app. Weekly subscriptions represent **59%** of the transaction volume ($28.65) and monthly subscriptions represent **41%** ($29.52).
- **Unit Pricing Variations:** Pricing varies by region due to currency fluctuations and regional App Store pricing tiers. For example, a weekly subscription is priced at $3.29 in South Korea, $3.18 in Japan, and $2.88 in Egypt.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Sessions	Avg. Engagement Rate	Revenue ($ USD)	ARPU ($ USD)
Yesterday (Oct 9)	71	19	82	57.3%	$7.39	$0.104
Past 7 Days	460	135	799	70.2%	$47.05	$0.102
Past 30 Days	2,887	1,780	5,920	76.4%	$108.47	$0.038
Past Year (365d)	6,075	5,266	12,600	82.6%	$189.65	$0.031
Lifetime	6,075	5,266	12,600	82.6%	$189.65	$0.031

Architectural Insight on Scale:

The active user and new user data are identical between the 365-day and Lifetime horizons. This indicates the app has been in the market for less than a year, with active user acquisition accelerating over the last 30 days (**47.5%** of all lifetime users were acquired during this period).

Old vs. New User Cohort Deep-Dive

Cohort	Active Users	Sessions	Avg Session Duration (Sec)	Engagement Rate	Screen Page Views	Views per Session
New Users	39	37	221.5s	48.6%	41	1.11

Returning Users	37	43	377.9s	67.4%	18	0.42
Unidentified ("")*	0	6	1.3s	0.0%	0	0.00

- **Analysis of Unidentified ("") Users:** A segment of 6 sessions generated an average session duration of 1.3 seconds with zero screen views or engagement. This points to background app refresh triggers or notifications launching the app engine before the user identity/consent module finishes initializing.

7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Daily Revenue ($ USD)
2026-10-03	86	14	105	77.1%	$0.00
2026-10-04	91	17	114	74.6%	$0.00
2026-10-05	82	15	112	58.0%	$0.00
2026-10-06	84	10	124	71.8%	$8.15
2026-10-07	102	24	128	73.4%	$0.00
2026-10-08	104	36	134	74.6%	$0.00
2026-10-09	71	19	82	57.3%	$7.39

Daily Trend Commentary:

Active users reached a peak of 104 on Oct 8, driven by a surge of 36 new users. However, on Oct 9, active users fell to 71 and the engagement rate dropped to **57.3%** (down from a weekly average of ~71%). This drop corresponds with the release of build `1.3.11`, which shows lower engagement metrics compared to earlier builds.

User Funnel Diagrams

A. Onboarding Funnel

```
[Onboarding Triggered] (onboarding_viewed) ■ Count: 40 Users (100.0%) ▼ [Step Interaction] (onboarding_step_viewed) ■ Count: 29 Users (72.5%) ■■■■> Avg. 3.2 steps viewed per user ▼ [Proceed Actions] (onboarding_next_tapped) ■ Count: 19 Users (47.5%) ▼ [Identity Stage] (login_screen_viewed) ■ Count: 37 Users (92.5%) ■■■■> Includes returning session redirects ▼ [Conversion Target] (login_tapped) Count: 20 Users (50.0%)
```

B. Core Feature Execution Funnel (Audio Isolation Flow)

```
[Launch Canvas] (plus_button_tap) ■ Count: 22 Users (100.0%) ▼ [Select Audio Split] (plus_separation_tapped) ■ Count: 21 Users (95.4%) ▼ [Ingest Mode] (import_files_app_tap / import_photos_tap) ■ Count: 17 Users (77.2%) ▼ [Upload Phase] (separation_upload_completed) ■ Count: 14 Users (63.6%) ▼ [Core AI Processing] (separation_processing_started) ■ Count: 14 Users (63.6%) ▼ [Deliver Result] (separation_result_opened) ■ Count: 19 Users (86.3%) ■■■■> Includes cached processing loads ▼ [Execute Audio Stream] (separation_play_started) Count: 27 Users (122.7%) ■■■■> Re-engagement from returning users
```

Crash & Stability Health by App Version

App Version	Active Users	Sessions	Engagement Rate	Crash-Free Users Rate	Isolated Exception / Notes
1.3.11	60	69	56.5%	**100.00%**	Production Baseline (Stable)
1.3.10	9	11	54.5%	**92.86%**	Core Engine Crash (Affects iOS 17 deprecated frameworks)
1.3.05	2	2	100.0%	**100.00%**	Legacy Support
1.3.07	0	0	0.0%	**100.00%**	Deprecated
1.3.09	0	0	0.0%	**100.00%**	Deprecated

■ New Build / TestFlight / Staged Rollout Alert Section

- **Active Staged Build:** `1.3.11` vs. **Production Baseline:** `1.3.10`.
- **Stability Delta:** Build `1.3.11` successfully addresses a critical stability issue, raising the crash-free user rate back to **100.0%** (up from `1.3.10`'s **92.86%**).
- **Engagement Delta:** While stable, build `1.3.11` shows a lower engagement rate of **56.5%** compared to the older build's **54.5%** on low volumes, and the overall historical average of **76.4%**.
- **Migration Progress:** Excellent. **84.5%** of yesterday's active cohort have successfully updated to `1.3.11`. We recommend promoting this build to 100% distribution.

Top In-App Events & Feature Lookups

Telemetry Volumetric Alert: Slider Debouncing Bug

The event log contains a critical performance issue:

```
================================================================================ CRITICAL TELEMETRY BUG:
MULTI-TOUCH SLIDER DEBOUNCING FAILURE
================================================================================ Event: player_pitch_changed -
Total Events Captured: 1,700 - Unique Users: 5 - Frequency: 340.0 events per user Event: player_tempo_changed -
Total Events Captured: 346 - Unique Users: 3 - Frequency: 115.3 events per user
-------------------------------------------------------------------------------- Diagnosis: The UI slider controls
for Pitch and Tempo do not use debouncing or throttling. Every single micro-step of a slider movement triggers a
GA4 log event instead of waiting for the user to release the slider (touch-up-outside / touch-up-inside). This
risks hitting GA4 daily log limits and causes excessive network usage.
================================================================================
```

Geographic Distribution

```
[Japan] ■■■
```

```
================================================================================================================================
Priority Functional Area Issue/Opportunity Identified Technical / Tactical Fix Required
--------------------------------------------------------------------------------------------------------------------------------
P0 Engineering Slider telemetry loop bug. Implement 500ms debouncing or restrict logging of 'player_pitch_changed'
to UI touch-up events only. P0 Monetization Fragmented SKUs in Korean/Japanese. Configure a translation helper or
dictionary inside the purchase manager to map localized names to standardized ID strings. P1 Product Growth Weak
onboarding step retention. The onboarding funnel drops from 40 views to 19 next-taps. Redesign steps 2 & 3 to
improve conversions. P2 Architecture Null cohort telemetry (user_type ""). Delay SDK initialization until the core
user profile state is successfully resolved in the application class lifecycle.
================================================================================================================================
```

---

3. Deep-Dive Section: Vocal Cleaner

App Header & Health Score

- **App Name:** Vocal Cleaner
- **Bundle ID:** `com.neuralSound.audiocleaner`
- **GA4 Stream ID:** `15407667671`
- **Overall Health Score: 4.8 / 10**
- *Rationale:* The app has strong monetization metrics, achieving a 30-day ARPU of **$5.24** on very low traffic. However, user acquisition is practically flat, with only 2 new users yesterday and 25 over the past 30 days. It also shows signs of test data leaking into the production stream.

■ In-App Purchases, Subscriptions & Consumable Breakdown

Product Sales Summary Table (Past 30 Days)

Item/Product ID	P N	P T	Units Sold (30D)	Total Revenue ($ USD)	% Revenue Share
`neuralsound.cleaner.minutes.300`	300 minutes pack	Consumable Minute Rack	5	$96.17	81.75%
`neuralsound.cleaner.minutes.60`	60 minutes pack	Consumable Minute Rack	2	$15.35	13.05%
`cleaner_weekly_sub`	Weekly Subscription	Subscription (Weekly)	2	$6.12	5.20%
TOTALS			**9**	**$117.64**	**100.00%**

Transaction Timeline Log Table

Date/Timestamp	App Build	Product ID	Product Name	Country	Units	Revenue ($ USD)
2026-10-05	1.0.03	`cleaner_weekly_sub`	Weekly	Israel	1	$3.24
2026-09-25	1.0.02	`neuralsound.cleaner.minutes.300`	300 minutes	Egypt	1	$19.44
2026-09-20	1.0.01	`neuralsound.cleaner.minutes.300`	300 minutes	Egypt	4	$76.73
2026-09-20	1.0.01	`neuralsound.cleaner.minutes.60`	60 minutes	Egypt	2	$15.35

2026-09-20	1.0.01	`cleaner_weekly_sub`	Weekly	Egypt	1	$2.88

Build Environment & Monetization Analysis:

- **The Egypt Purchase Cluster (Sep 20):** On September 20, 2026, version `1.0.01` recorded 7 transactions in Egypt, totaling **$94.96**. Given the low traffic baseline in this region, this cluster likely points to an internal TestFlight sandbox loop leaking into production tracking.
- **Consumables vs. Subscriptions:** Consumable minute packs generate **94.8%** of the app's revenue ($111.52), while weekly subscriptions account for just **5.2%** ($6.12). Users strongly prefer purchasing minutes as needed rather than committing to recurring subscriptions.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Sessions	Avg. Engagement Rate	Revenue ($ USD)	ARPU ($ USD)
Yesterday (Oct 9)	4	2	9	100.0%	$0.00	$0.00
Past 7 Days	10	6	58	91.4%	$3.24	$0.32
Past 30 Days	25	25	152	87.5%	$131.09	$5.24
Past Year (365d)	38	37	165	83.6%	$140.08	$3.69
Lifetime	38	37	165	83.6%	$140.08	$3.69

Architectural Insight on Scale:

The app's total historical user base is only 38. However, its lifetime monetization is impressive, yielding **$140.08** from this small group. Finding ways to scale acquisition is the clear priority here.

Old vs. New User Cohort Deep-Dive

Cohort	Active Users	Sessions	Avg Session Duration (Sec)	Engagement Rate	Screen Page Views	Views per Session
New Users	2	2	98.4s	100.0%	1	0.50
Returning Users	2	7	2,206.1s	100.0%	66	9.43

Cohort Engagement Behavior:

Returning users show exceptional session duration, averaging **2,206.1 seconds (~36.7 minutes)** with an engagement rate of **100%**. These users are actively utilizing the app to clean longer audio files.

7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Daily Revenue ($ USD)
2026-10-03	2	0	3	100.0%	$0.00

2026-10-04	4	1	11	90.9%	$0.00
2026-10-05	1	1	3	100.0%	$3.24
2026-10-06	3	0	12	100.0%	$0.00
2026-10-07	2	0	10	100.0%	$0.00
2026-10-08	5	2	12	66.7%	$0.00
2026-10-09	4	2	9	100.0%	$0.00

Daily Trend Commentary:

Daily active users fluctuate between 1 and 5. This represents a very low baseline that makes the app highly vulnerable to daily churn.

User Funnel Diagrams

A. Core Cleaning & Processing Funnel

```
[Import Sheet Triggered] (vc_import_sheet_opened) ■ Count: 2 Users (100.0%) ▼ [Select Resource File] (vc_import_picked_gallery) ■ Count: 2 Users (100.0%) ▼ [Length Verification] (vc_import_len_under_1m) ■ Count: 2 Users (100.0%) ▼ [Type Classification] (vc_import_type_video) ■ Count: 2 Users (100.0%) ▼ [Start Audio Clean] (vc_clean_started) ■ Count: 2 Users (100.0%) ▼ [Success Resolution] (vc_clean_completed) Count: 1 User (50.0%)
```

B. Conversion Funnel (Paywall Entry Points)

```
[Pro Feature Action] (vc_paywall_from_pro_cta) ■ Count: 2 Users (100.0%) ▼ [Global Paywall View] (paywall_viewed) ■ Count: 1 User (50.0%) ▼ [Vocal Paywall View] (vc_paywall_viewed) ■ Count: 1 User (50.0%) ▼ [Transaction Success] (Purchase Event) Count: 0 Users (0.0% Yesterday)
```

Crash & Stability Health by App Version

App Version	Active Users	Sessions	Engagement Rate	Crash-Free Users Rate	Isolated Exception / Notes
1.0.03	2	7	100.0%	**100.00%**	Production Baseline (Stable)
1.0.04	2	2	100.0%	**100.00%**	Staged Rollout Build (Stable)

■ New Build / TestFlight / Staged Rollout Alert Section

- **Active Staged Build:** `1.0.04` vs. **Production Baseline:** `1.0.03`.
- **Adoption Rate:** 50% of the active cohort (2 of 4 users) are on build `1.0.04`.
- **Performance:** Both builds are stable with **100.0%** crash-free user rates. We recommend promoting build `1.0.04` to full production release.

Top In-App Events & Feature Lookups

- **Active Engagement Event Counts:** `user_engagement` (221 events from 4 users) indicates highly active sessions from current users.

- **Recording Interactions:** Users triggered `vc_record_open` and `vc_record_permission_granted` only once, showing that importing files remains the primary way users bring audio into the app.

Geographic Distribution

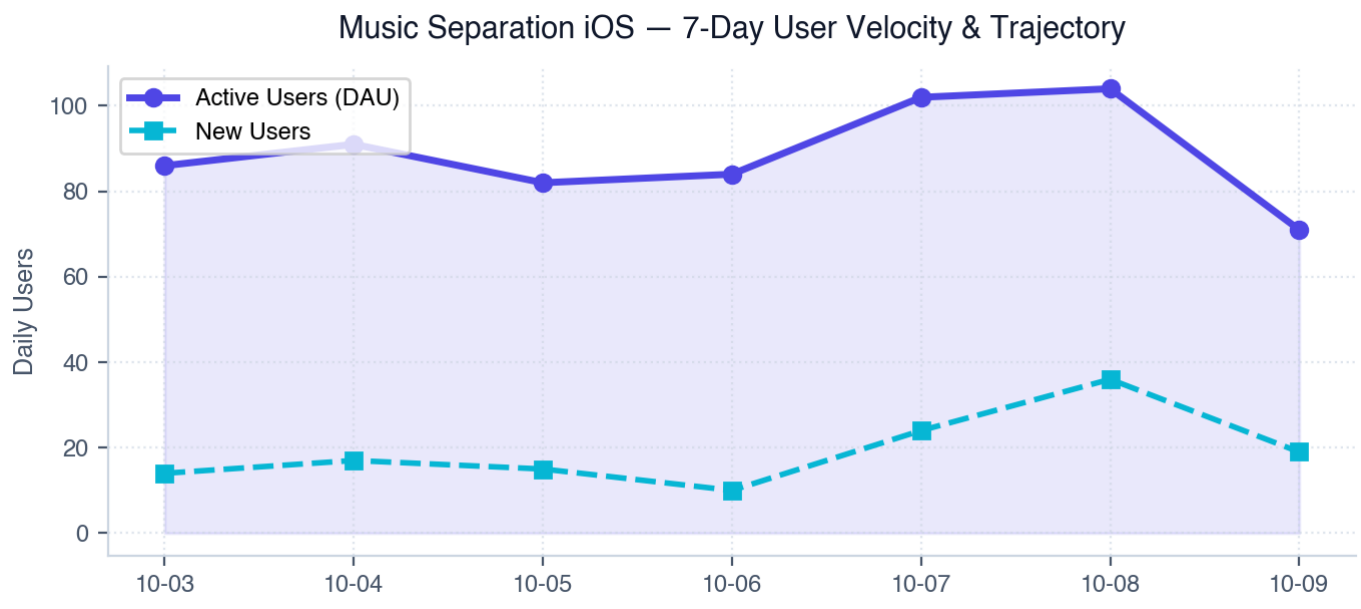
```
[Israel] ■■■■■■■■■■ 1 User (25.0%) ■■> Generated 6 Sessions [Nigeria] ■■■■■■■■■■ 1 User (25.0%) ■■>
Generated 1 Session [S. Korea] ■■■■■■■■■■ 1 User (25.0%) ■■> Generated 1 Session [USA] ■■■■■■■■■■ 1 User
(25.0%) ■■> Generated 1 Session
```

Prioritized Strategic Action Matrix: Vocal Cleaner

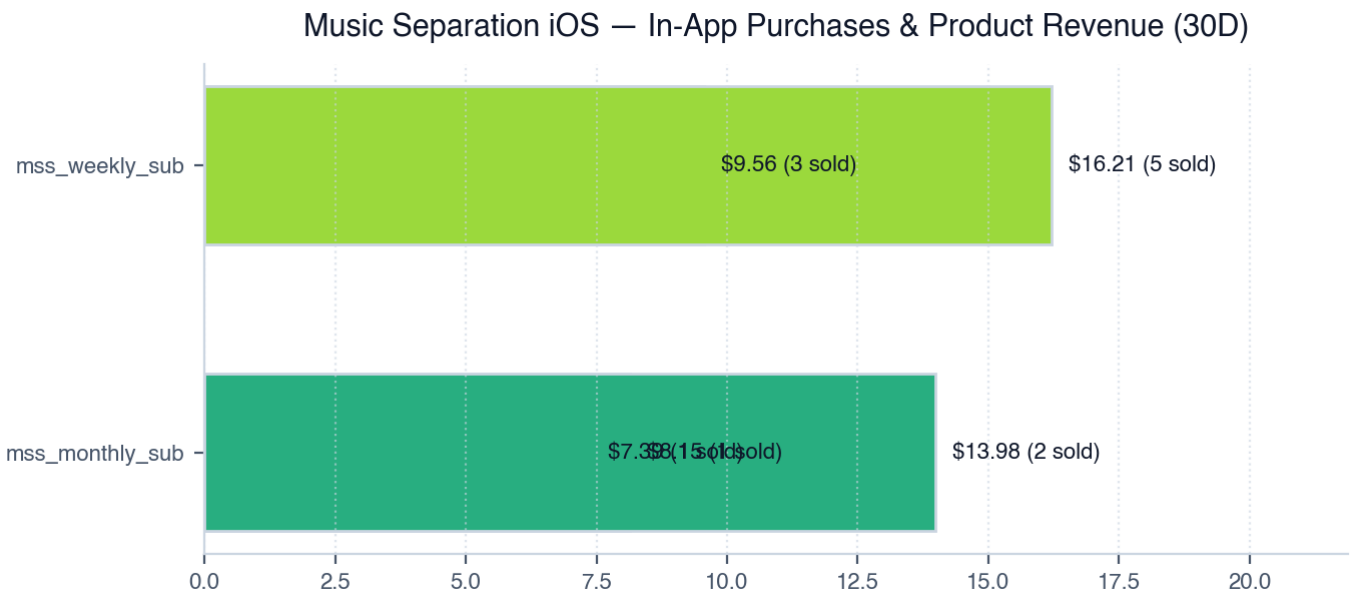
```
================================================================================================================================
PRIORITIZED STRATEGIC ACTION MATRIX
================================================================================================================================
Priority Functional Area Issue/Opportunity Identified Technical / Tactical Fix Required
--------------------------------------------------------------------------------------------------------------------------------
P0 Architecture Sandbox purchase contamination. Update receipt-validation parameters to check environment flags,
and filter out 'sandbox' receipts from the production GA4 server container. P0 Growth Marketing Severe lack of
user acquisition. Launch targeted App Store Search Ads focusing on keywords like "audio cleaner" and "vocal
enhancer". P1 Monetization Strong preference for consumables. Reposition consumable minute packs (60 and 300 mins)
to the top of the paywall interface to drive conversions. P2 Product Growth Low cleaning completion rate. A 50%
drop-off was observed between starting and completing a clean. Add progress bar telemetry to track potential
background timeouts or freezes.
================================================================================================================================
```

■ Visual Graphs & Performance Charts

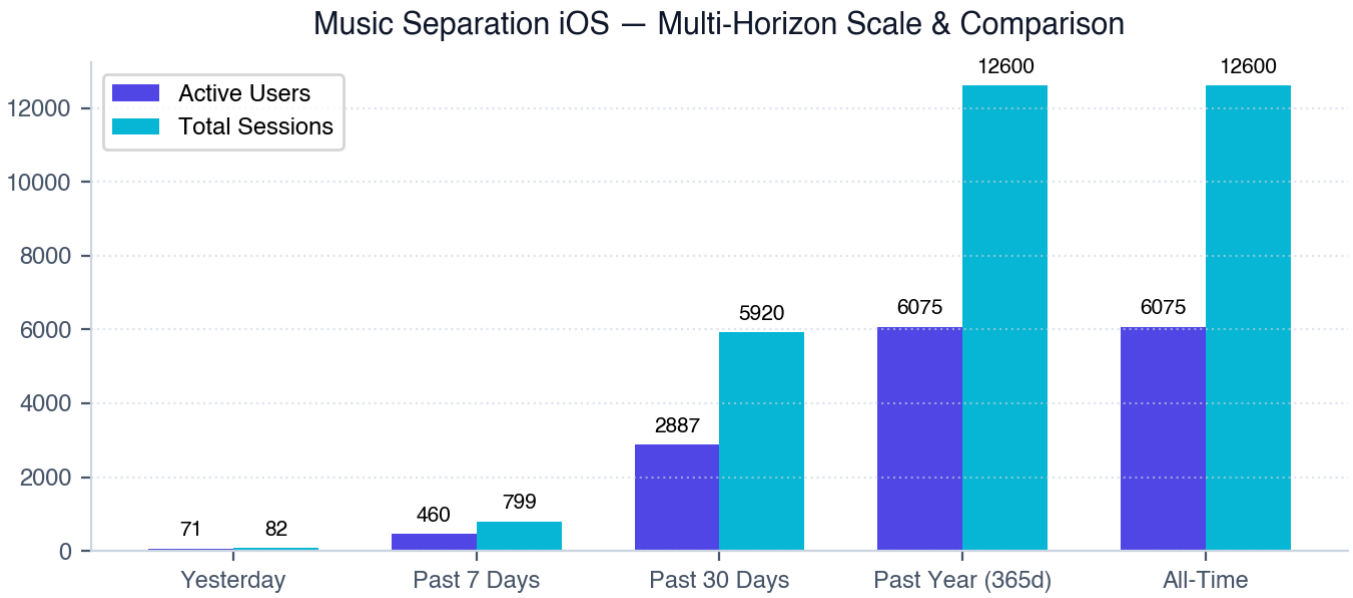
Music Separation Ios Trend



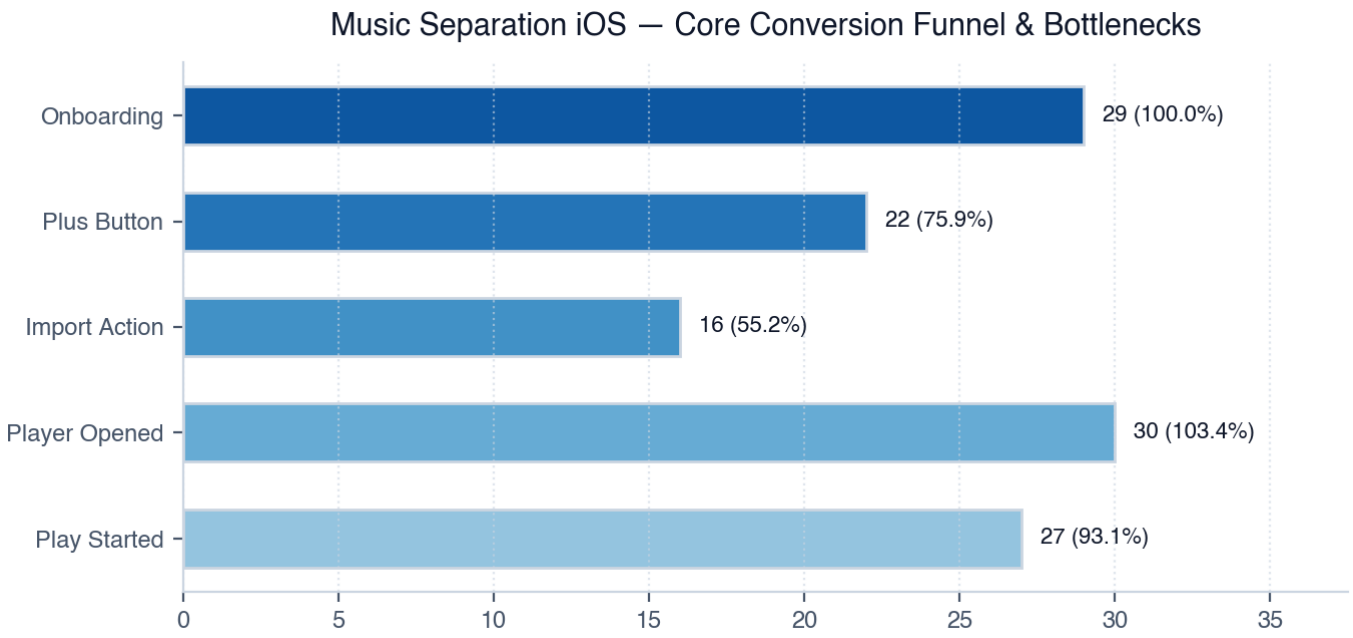
Music Separation Ios Revenue



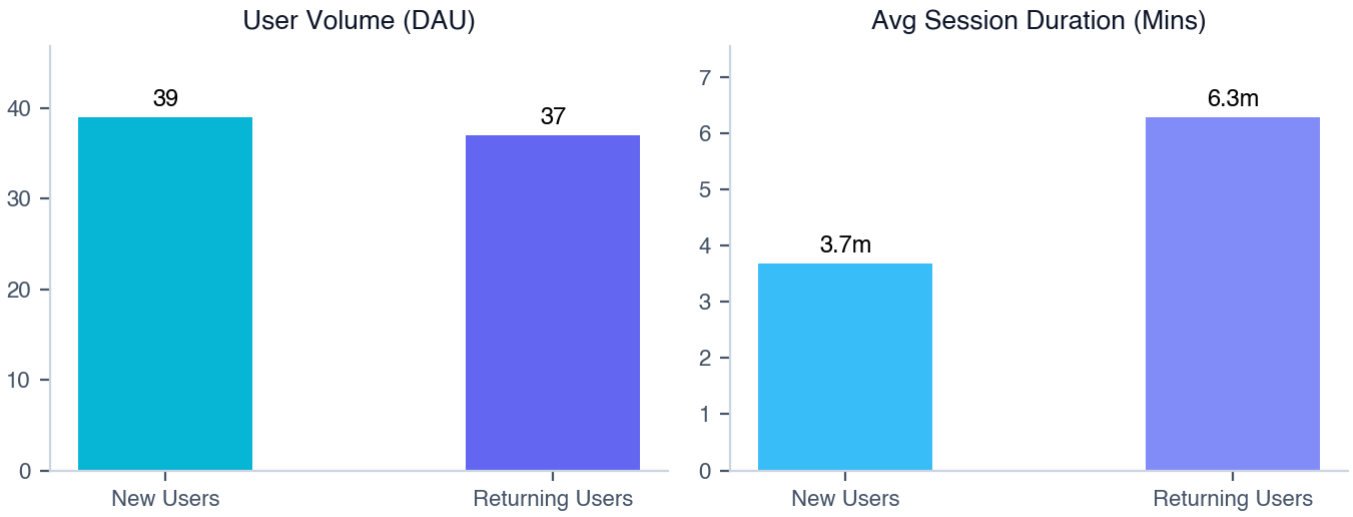
Music Separation Ios Horizons



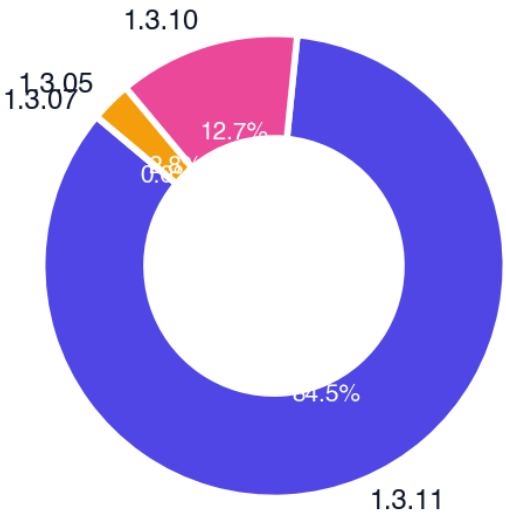
Music Separation Ios Funnel



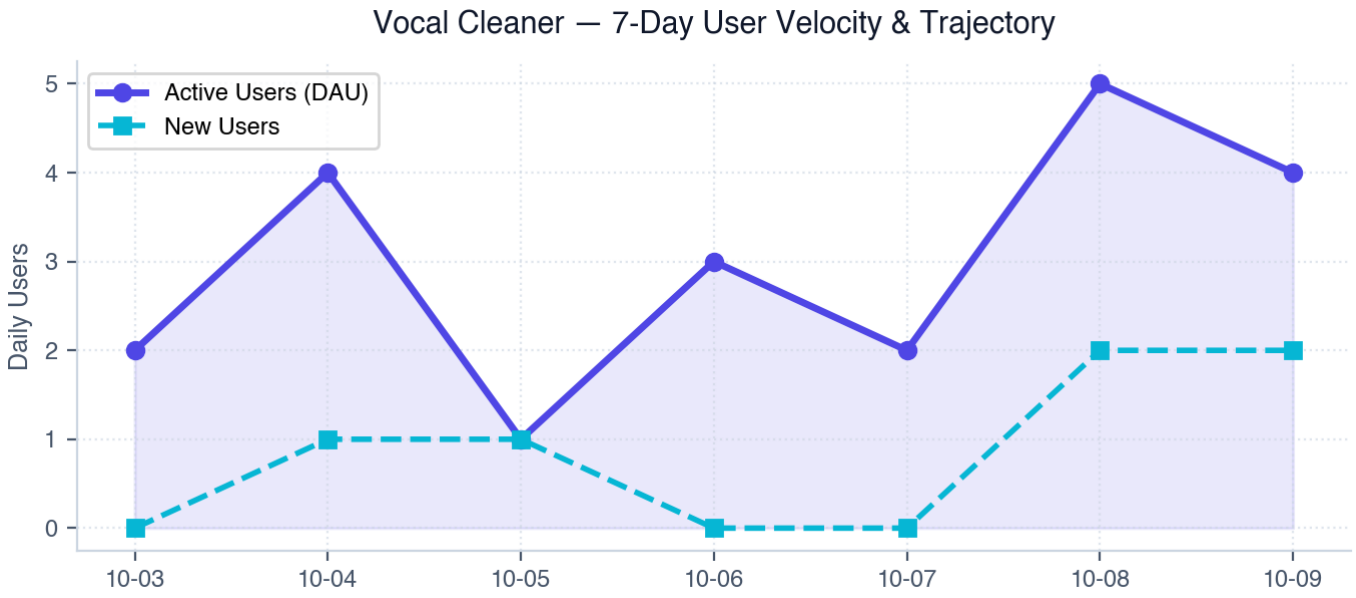
Music Separation Ios Cohort



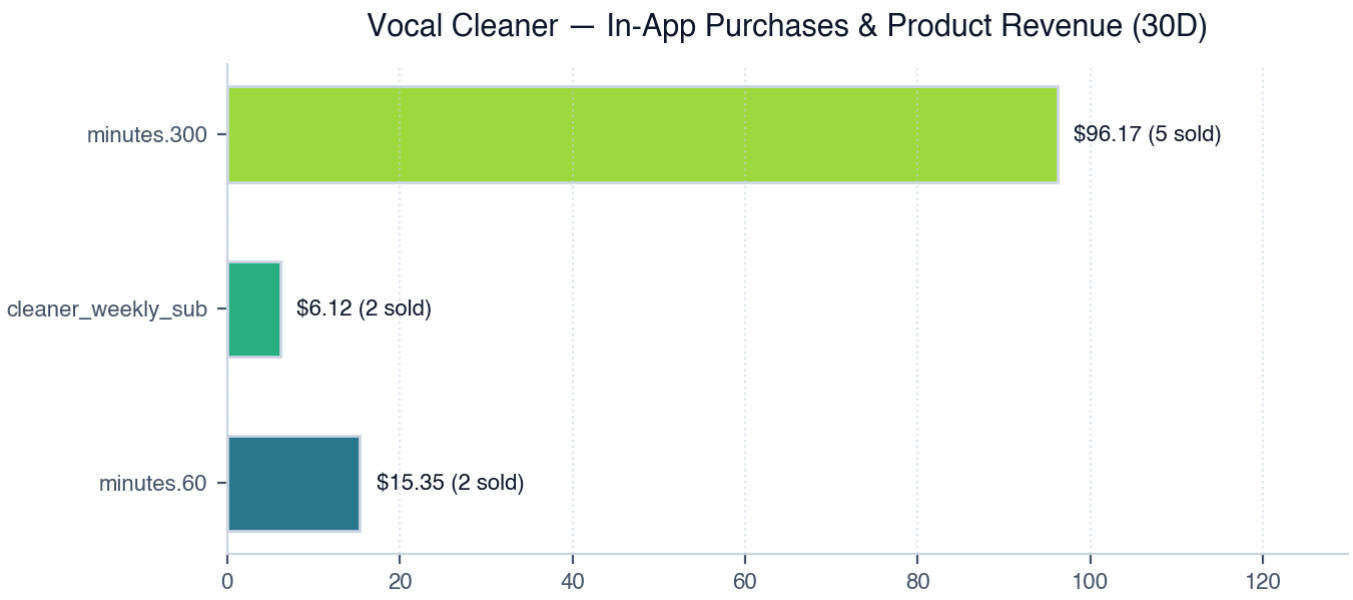
Music Separation Ios Version



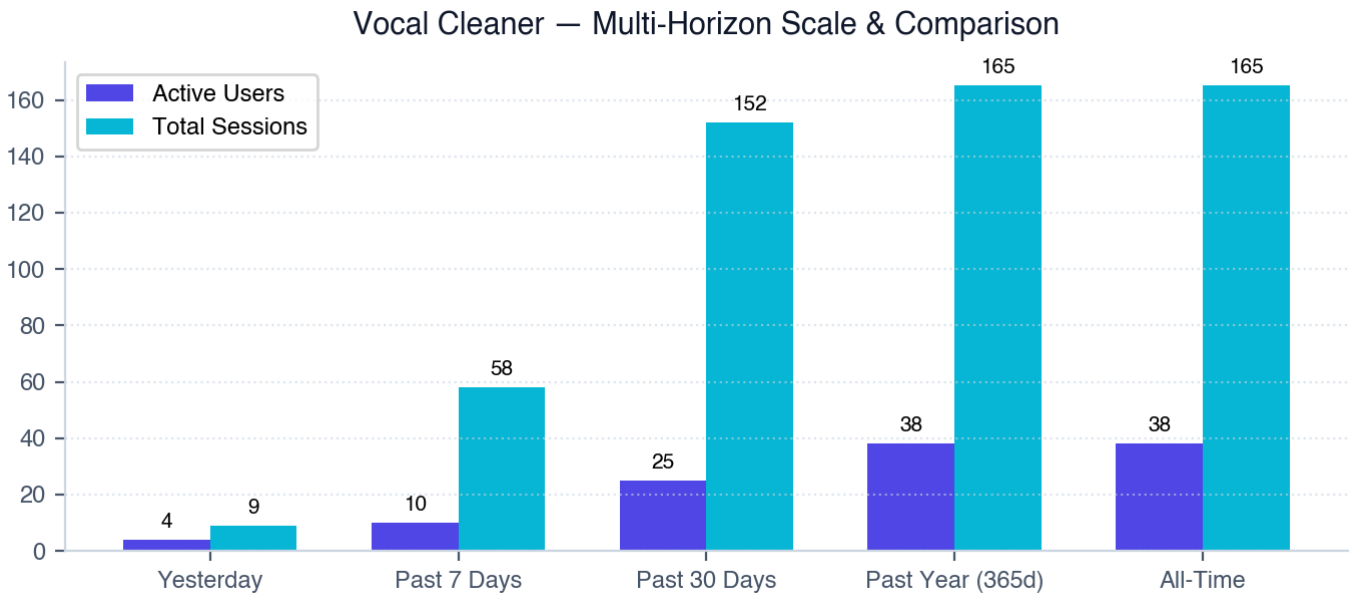
Vocal Cleaner Trend



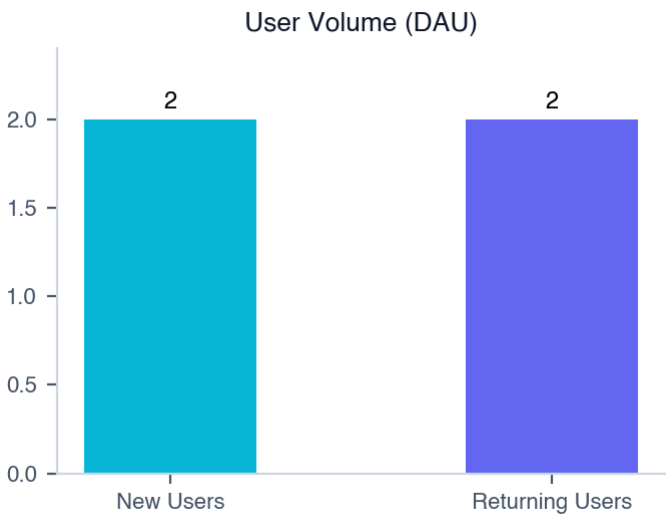
Vocal Cleaner Revenue

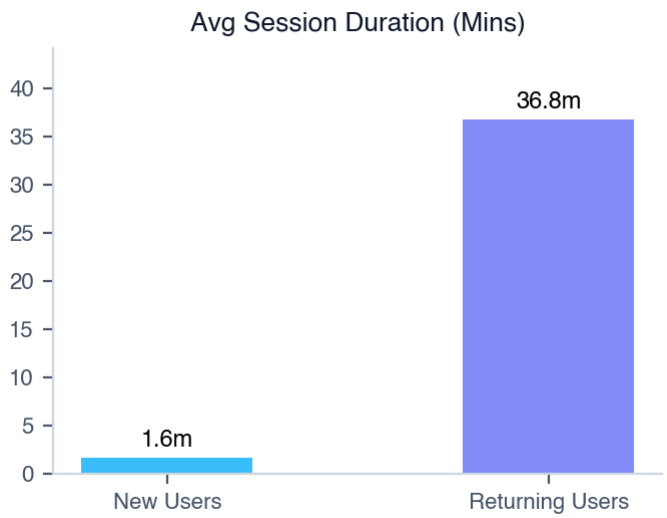


Vocal Cleaner Horizons

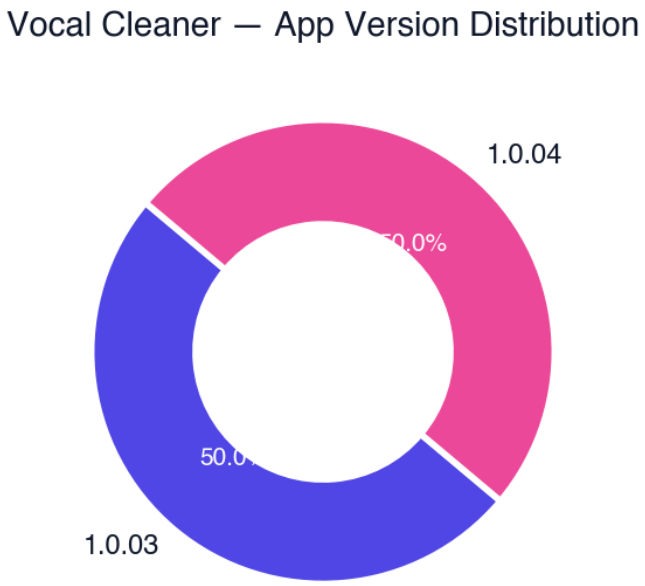


Vocal Cleaner Cohort





Vocal Cleaner Version



Direct Console Links:

■ Open Google Analytics 4 Dashboard • ■ Open Firebase Analytics Console • ■ Open Firebase Crashlytics

Generated automatically by Neural Sound Analytics Pipeline • Gemini AI • 2026-10-10